

# INTEGER OPTIMIZATION NEVER DIES

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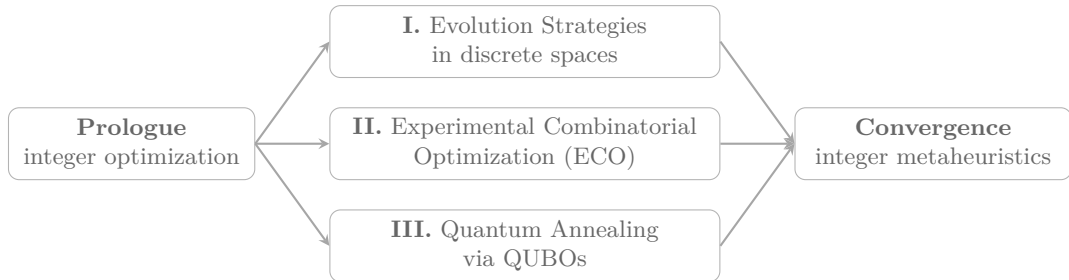
**TEL-HAI**

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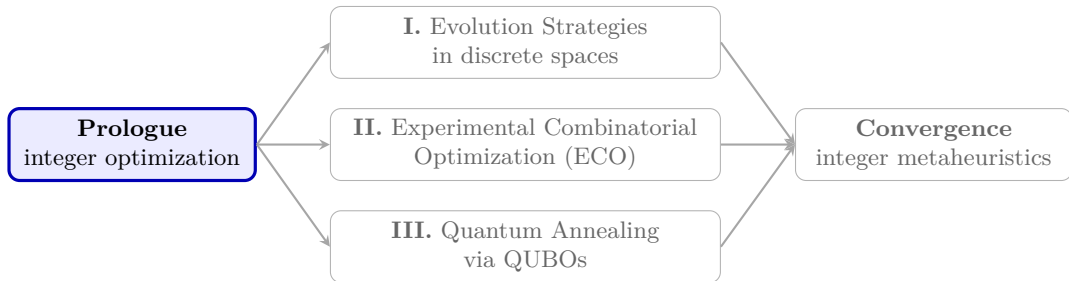
**U-Exeter CS Seminar, 6 October 2026**

# the talk in one slide

*Integer optimization is not a “solved” classical relic — it is undergoing a renaissance, fueled by non-classical solvers and a shift toward experimental methodologies.*



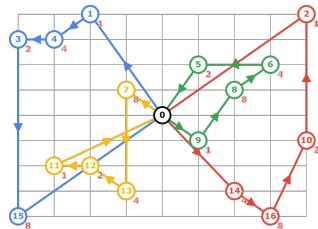
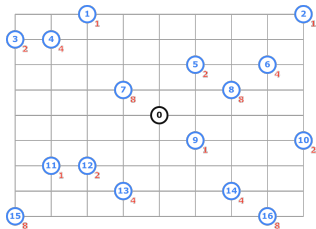
## roadmap





(business) motivation: supply-chain / logistics problems

# capacitated vehicle routing problem

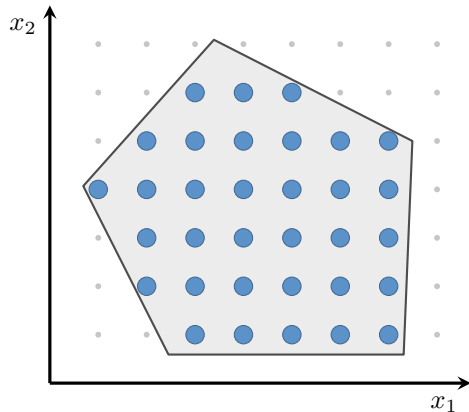


- vehicles of limited capacity deliver items to customers at the least total routing cost (Dantzig & Ramser, 1959)
- contains the **TSP** (routing) and **bin packing** (capacity)  $\Rightarrow \mathcal{NP}$ -hard
- **state of the art**: exact branch-cut-and-price proves optimality up to a few hundred customers; dozens of benchmark instances remain open (Uchoa et al., 2017).  
Beyond that, **heuristics** rule — e.g., Hybrid Genetic Search (Vidal, 2022)

## mathematical formulation

$$\begin{array}{ll}
 \text{minimize}_{\vec{x}} & f(\vec{x}) \quad \vec{x} \in \mathbb{Z}^d \\
 \text{subject to:} & g_k(\vec{x}) \geq 0 \quad k = 1, \dots, m \\
 & h_j(\vec{x}) = 0 \quad j = 1, \dots, \ell
 \end{array}$$

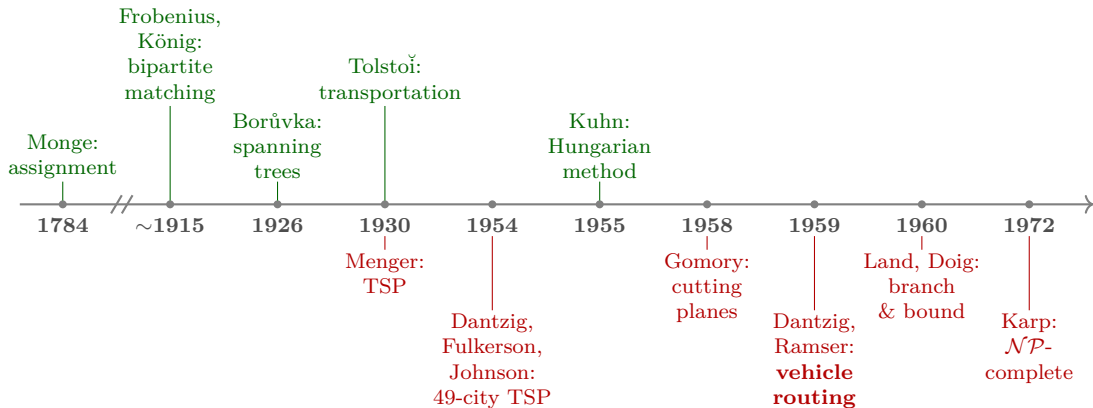
- Optimization is concerned with finding **global minima** of a function, e.g.,  $f$  above.
- In engineering: finding the best system design subject to **constraints** — inequalities  $g_k$  and equalities  $h_j$ .
- Integer optimization: the feasible points are the **lattice points** of  $\mathbb{Z}^d$  that satisfy them.



feasible points: the lattice  $\mathbb{Z}^d$  inside the region

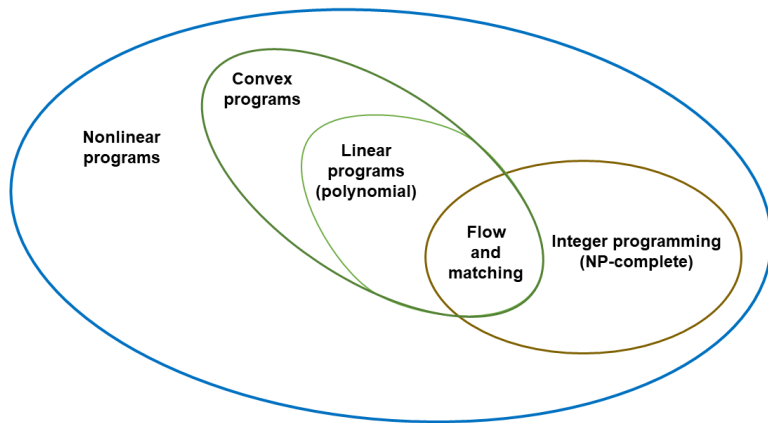
# roots of combinatorial optimization

The roots are *easy* problems; *hardness* — and IP methods — arrive in the 1950s:



A. Schrijver, “On the history of combinatorial optimization (till 1960)”, 2005.

## taxonomy

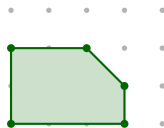


Source: “Combinatorial Optimization” by Ch. Papadimitriou & K. Steiglitz.

## when is integrality for free?

ILP:  $\max\{\vec{c}^\top \vec{x} : A\vec{x} \leq \vec{b}, \vec{x} \in \mathbb{Z}^d\}$ , with  
 $P = \{\vec{x} \in \mathbb{R}^d : A\vec{x} \leq \vec{b}\}$ ,  $P_I = \text{conv}(P \cap \mathbb{Z}^d)$ .

- $P = P_I \Rightarrow$  the **LP relaxation** solves the ILP in polynomial time.
- **Total unimodularity** (Hoffman & Kruskal, 1956):  
 all square submatrices of  $A$  have  $\det \in \{0, \pm 1\} \Leftrightarrow$   
 $\{\vec{x} \geq \vec{0} : A\vec{x} \leq \vec{b}\}$  integral for all integral  $\vec{b}$   
 e.g., assignment, bipartite matching, flows, transportation,  
 shortest paths
- **Known integral hull:** general matching is not TU,  
 yet Edmonds (1965) gave  $P_I$  — exponentially many  
 odd-set inequalities, separable in polynomial time.

 $P = P_I$  (TU) $P \neq P_I$ : fractional vertex

Easy  $\Leftrightarrow$  optimize over  $P_I$  efficiently  $\Leftrightarrow$  separate over  $P_I$  efficiently (Grötschel, Lovász & Schrijver, 1981).

# the knapsack problem

$d$  items, picked entirely (not fractionally):

- $v_i$ : value,  $w_i$ : weight
- capacity  $C$ ; **maximize the total value**

$$\begin{array}{ll}
 \text{[KP]} & \text{maximize } \sum_{i=1}^d x_i \cdot v_i \\
 & \text{subject to:} \\
 & \sum_{i=1}^d x_i \cdot w_i \leq C \\
 & x_i \in \{0, 1\} \quad \forall i \in 1 \dots d
 \end{array}$$

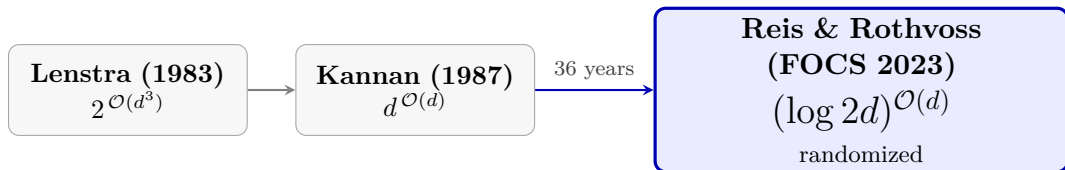
**A single constraint row — yet hard:**

- $A = \vec{w}^\top$  is **not TU** (any  $w_i \geq 2$ )
- $P \neq P_I$ : the LP optimum is greedy by  $v_i/w_i$  and leaves **one fractional item** (Dantzig, 1957)
- describing  $P_I$  needs families such as **cover inequalities**

KP is only weakly  $\mathcal{NP}$ -hard: dynamic programming solves it in  $\mathcal{O}(dC)$ , and a  $(1 - \epsilon)$ -approximation is computable in time polynomial in  $d$  and  $1/\epsilon$ . The multidimensional KP ( $m$  constraints,  $m$  part of the input) is strongly  $\mathcal{NP}$ -hard.

# integer programming: a recent breakthrough

The decision version is  $\mathcal{NP}$ -complete (Karp, 1972); unbounded IP with quadratic constraints is even undecidable (Jeroslow, 1973).

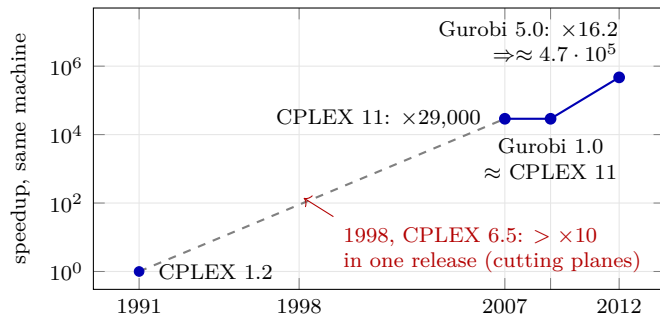


running times in the dimension  $d$ , up to a polynomial in the input size: all three are polynomial for fixed  $d$

Conjectured by Kannan & Lovász (1988); proved up to a polylogarithmic factor.

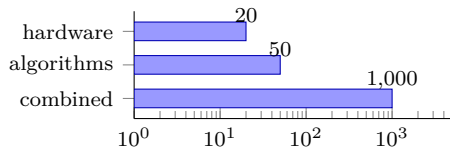
**Still exponential in  $d$  — heuristics remain the primary problem-solvers.**

# MIP solvers: three decades of speedups



**Same machine (geom. mean):**  
a 1-year solve in 1991  $\rightarrow$  about **1 minute** in 2012.

**MILP, 2001–2020:**



**By the way:** CPLEX's polish heuristic is an **evolutionary algorithm** — it mutates and recombines the solutions found so far, completing each offspring by a sub-MIP (Rothberg, INFORMS J. Comput. 19, 2007; in CPLEX since v10, 2006)

R. E. Bixby, Doc. Math. Extra Vol. ISMP, 107–121, 2012 • T. Koch et al., EURO J. Comput. Optim. 10, 2022

# so why does integer optimization never die?

## Theory keeps advancing ...

- exact algorithms:  $d^{\mathcal{O}(d)} \rightarrow (\log 2d)^{\mathcal{O}(d)}$   
— but only on the linear/convex
- MIP solvers:  $\sim 10^5$ -fold algorithmic speedup, 1991–2012

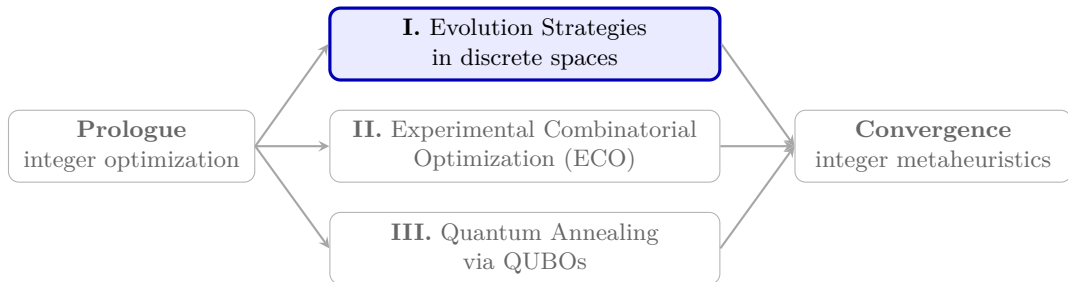
## ...yet practice is heuristic

- general integer programs: undecidable
- $\mathcal{NP}$ -hard instances at industrial scale
- heuristics remain the workhorses

## New drivers of a renaissance:

- ① **modern ES** – self-adaptive, small-population search, now with integer-native machinery
- ② **experiments in the loop** – objectives with *no* formula at all
- ③ **non-classical solvers** – quantum annealers natively minimize QUBOs

## roadmap



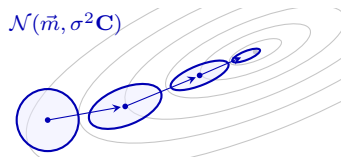
# evolution strategies: a success story in $\mathbb{R}^d$

One iteration of a  $(\mu/\mu_w, \lambda)$ -ES:

$$\vec{x}_k = \vec{m} + \sigma \vec{z}_k, \quad \vec{z}_k \sim \mathcal{N}(\vec{0}, \mathbf{C})$$

$$\vec{m} \leftarrow \sum_{i=1}^{\mu} w_i \vec{x}_{i:\lambda}$$

$$(\sigma, \mathbf{C}) \leftarrow \text{adapt}(\sigma, \mathbf{C}; \vec{x}_{1:\lambda}, \dots, \vec{x}_{\mu:\lambda})$$



**1960s–70s** Rechenberg and Schwefel, Berlin: the ES, the 1/5th success rule, self-adaptation

**1996–2001** **CMA-ES** (Hansen & Ostermeier): adapts the full covariance;  $\mathbf{C} \propto \mathbf{H}^{-1}$  on convex quadratics; the reference in continuous black-box optimization

**2008–2017** the theory catches up: the core update is a **natural-gradient** step on the Gaussian family (NES, IGO)

In  $\mathbb{R}^d$  the heuristic came first and the mathematics followed: the Gaussian is the maximum-entropy model, and its adaptation follows the natural gradient.

Rechenberg, 1973 • Schwefel, 1981 • Hansen & Ostermeier, 2001 • Wierstra et al., 2014 • Ollivier et al., 2017

then came the integers: round, repair — and it works



### The practice

- rounding alone **stalls**: for  $\sigma$  well below the lattice spacing, the samples round to the same point
- hence the **repairs**: integer mutations, variance bounds, probability margins
- and it **works**: the repaired CMA-ES is the state of the art among integer ESs

### The questions no one asked

- which mutation distribution on  $\mathbb{Z}^d$ ?
- what is a “step size” on a lattice?
- what does the adaptation estimate?

A singular exception: Rudolph (1994), maximum entropy  $\Rightarrow$  DG mutation; its step-size control stayed heuristic.

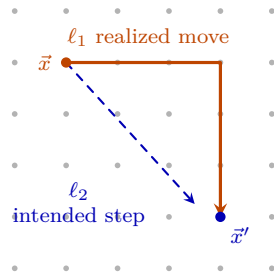
It works — but no one tried to make it mathematically sound. **Until now.**

## integer lattices change the local geometry

Continuous search is guided by  $\ell_2$  geometry; on the integer lattice, moves are coordinate-wise ( $\ell_1$ ):

$$\|\vec{x} - \vec{x}'\|_2 \longrightarrow \|\vec{x} - \vec{x}'\|_1 = \sum_{i=1}^d |x_i - x'_i|$$

- **continuous mutation does not stay continuous:**  
a sample in  $\mathbb{R}^d$  is off the lattice — it must be projected or **rounded**
- equal Euclidean length, different lattice length:  $(2, 0)$  and  $(1.2, 1.6) \mapsto (1, 2)$  both have  $\|\cdot\|_2 = 2$ , but  $\ell_1$  lengths 2 and 3



**Why this matters:** after rounding, the realized move is no longer the intended step.

# mutation: rounded Gaussian versus lattice-native DG

Continuous ESs use Gaussian steps naturally. In  $\mathbb{Z}^d$ , the reflex is to round (truncate); DG instead samples integer steps directly (Rudolph, 1994).

## Truncated Normal (TN)

- draw in  $\mathbb{R}^d$ , project to  $\mathbb{Z}^d$
- rounding changes the probabilities
- scaled through  $\sigma$  — not a lattice length

$$\Pr\{\text{rnd}(z) = k\} = \Phi\left(\frac{k+1/2}{\sigma}\right) - \Phi\left(\frac{k-1/2}{\sigma}\right)$$

## Double Geometric (DG)

- draw directly on  $\mathbb{Z}^d$
- maximum entropy for a fixed  $\mathbb{E}|Z|$
- scaled through  $\delta := \mathbb{E}|Z|$  — the expected lattice step

$$\Pr(Z = k) = \frac{1-q}{1+q} q^{|k|}, \quad \delta = \frac{2q}{1-q^2}$$

**Key point:** DG is not a rounded continuous mutation; it is a **lattice-native** operator — an exponential family ( $\theta = \log q$ ) whose sufficient statistic  $|Z|$  is the lattice displacement.

# Integer Natural Evolution Strategies (INES)

**Research question.** *Can integer step-size adaptation be derived from first principles, using a native  $\ell_1$ -norm mutation model and information geometry?*

**DG mutation**  
(max-entropy)



**sufficient  
statistic**  
 $|z_i|$



**natural  
gradient**  
(Fisher)



**cumulation**  
(evolution path)



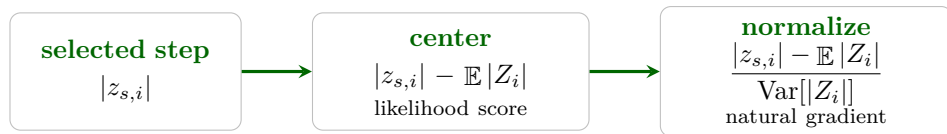
**$(1, \lambda)$ -INES**

**Reminder — NES** (Wierstra et al., 2014): maintain a search distribution  $\pi(\vec{z} \mid \theta)$  and minimize  $J(\theta) := \mathbb{E}_\theta [f(\vec{m} + \vec{z})]$  along the **natural gradient** (Amari, 1998):

$$\tilde{\nabla}_\theta J = \mathcal{F}(\theta)^{-1} \mathbb{E}_\theta [f(\vec{m} + \vec{z}) \nabla_\theta \log \pi(\vec{z} \mid \theta)] \approx \mathcal{F}(\theta)^{-1} \frac{1}{\lambda} \sum_{k=1}^{\lambda} f(\vec{m} + \vec{z}_k) \nabla_\theta \log \pi(\vec{z}_k \mid \theta)$$

the sufficient statistic gives the adaptation signal

For an exponential family, the likelihood score is the **observed sufficient statistic minus its expectation**; the Fisher information is  $\text{Var}[|Z_i|]$ .

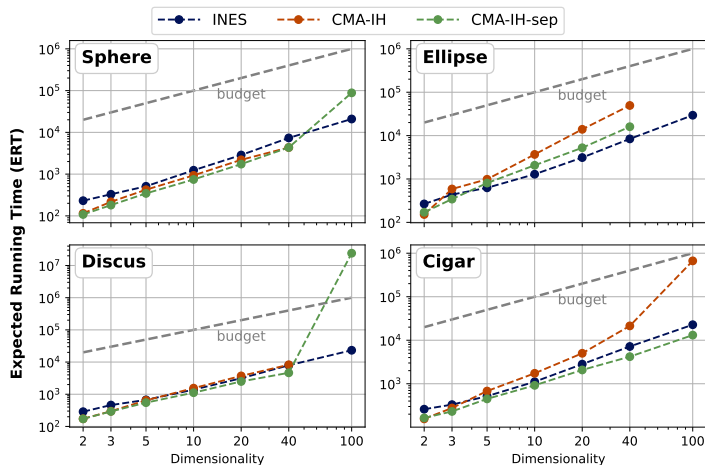


With the DG moments:

$$\tilde{\nabla}_{\theta_i} = \frac{|z_{s,i}| - \delta_i}{\delta_i \sqrt{1 + \delta_i^2}}$$

de Nobel, J., Vermetten, D., Wang, H., Shir, O.M., Emmerich, M., Bäck, T.: Integer Natural Evolution Strategies. In: Proceedings of Parallel Problem Solving from Nature (PPSN 2026). Lecture Notes in Computer Science, vol 16987. Springer, Cham (2026) 50-65

## benchmarking on unimodal quadratic functions



25 runs per function and dimension; budget  $10^4 \cdot d$ ; lower ERT is better. Baselines: **CMA-IH**, the dedicated integer-handling CMA-ES — default and separable (-sep).

- **Ellipse**: INES fastest from  $d = 5$  on
- $d = 100$ : INES solves all four; CMA-IH-sep exceeds the budget on Discus
- low  $d$ : INES pays a small constant factor

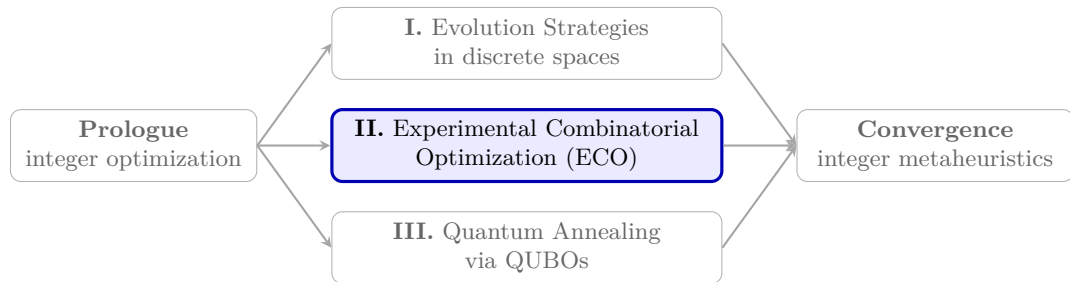
## part I: take-home

- integer ESs need not be rounded continuous ESs: the DG distribution gives a **lattice-native** mutation model
- its sufficient statistic  $|z_i|$  yields a **natural-gradient** signal for step-size adaptation
- cumulation turns this signal into a stable  $(1, \lambda)$ -INES update
- INES is most compelling on **ill-conditioned** (Ellipse) landscapes and robust in **high dimension**

**Next steps:** no global step size yet (coordinate-wise only) — multi-parent recombination, a global scale, and **dependencies between integer variables** (correlated mutations on  $\mathbb{Z}^d$ ).

**Bridge to Part II:** the same small-population, self-adaptive ES machinery runs inside our experimental loops — where every evaluation is a laboratory experiment.

## roadmap



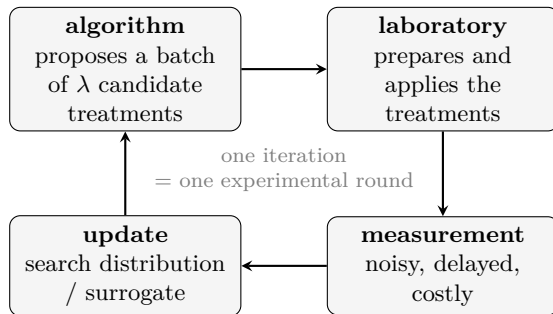
# experimental combinatorial optimization: zero presumption

## Theoretical / simulation-based $f$

- a formula or a simulator
- cheap, repeatable, deterministic
- evaluations in the millions

## Experimental $f$ (ECO)

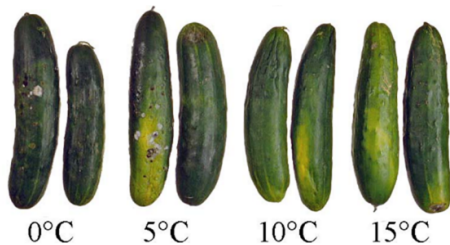
- **no model**: only a lab measurement
- costly, noisy, delayed, in batches
- **tens to hundreds** of evaluations



**Zero presumption:** nothing is assumed about the response beyond the decision space.

## case study: postharvest protocols for cucumbers

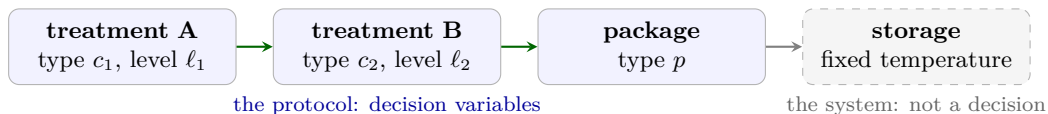
- **postharvest**: handling crops right after harvest, to maintain their quality and extend their shelf-life — a direct lever on *food security*
- **cucumbers** as a model: extremely sensitive, with a postharvest potential of under 14 days
  - rots, chilling injury, yellowing, shriveling, softening
- available all year round  $\Rightarrow$  **sequential experimentation**



blemishes vs. storage temperature

**Goal:** an optimized combination of **two treatments** and a **package** that minimizes the loss of quality during storage and marketing.

# the decision space: a mixed categorical–integer protocol



- $n_t = 10$  treatments: growth regulators, UV-C, volatiles, coatings, wax, ...
- each with integer **activation levels** (concentration, duration)
- 3 package types

Any order of the  $n_t$  treatments, then a package:

$$|\pi \circ \mathcal{T}| = n_t! \cdot \left[ \prod_{j=1}^{n_t} (\mathcal{T}_j^{\max} - \mathcal{T}_j^{\min}) \right] \cdot |\mathcal{P}| \approx 10^{17}$$

Industrial practice allows only two treatments:

$$\vec{\varphi} := \left( \underbrace{c_1, c_2, p}_{\text{categorical}}, \underbrace{\ell_1, \ell_2}_{\text{integer}} \right)^T \implies \approx 10^6 \text{ protocols}$$

20 °C and 10 °C are **independent systems**: two optimization problems in parallel.

the objective function is a laboratory measurement

$$\mathcal{L}_{i \rightarrow f}(\vec{\tau}) := \Delta c(\vec{\tau}) + \Delta s(\vec{\tau}) + \Delta m(\vec{\tau}) + \text{score}_{\text{exp}}(\vec{\tau}) \longrightarrow \min$$

deterioration from harvest to the end of storage: color deviation, stiffness deviation, mass reduction, and an expert's score (each normalized)

- measured after **4 weeks** in storage, with **10 biological repetitions**
- separately in each system: 20 °C (immediate marketing), 10 °C (cold storage); constraint: no decay



**Budget:** 7 generations  $\times$  (11 candidates + 2 references) per system — **77 evaluations** in a space of  $\sim 10^6$ . This is *not* a global optimization challenge. Strategy: a categorical ES.

## 20 °C system: outcome after 28 days



protocols proposed by the algorithm; **red frame**: the best outcome

**control** (untreated)



**“in-house”**  
(best human practice)



10 °C system: storage prolonged up to 63 days



control, 28 days



algorithm, 28 days



algorithm, **63 days**

Storage prolonged up to **63 days** at 10 °C — for a crop whose postharvest potential is under 14 days.

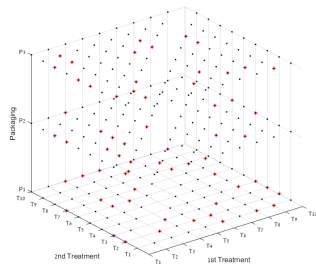
# lessons from the lab

## 1. How to compare across generations?

Next to its candidates ( $\mathcal{L}^{(g)}$ ), every generation  $g$  re-measures two references: control ( $\mathcal{L}_0^{(g)}$ ) and in-house ( $\mathcal{L}_{\text{ih}}^{(g)}$ ). Two normalizations were tested:

$$f := \frac{\mathcal{L}^{(g)} - \mathcal{L}_{\text{ih}}^{(g)}}{\mathcal{L}_0^{(g)} - \mathcal{L}_{\text{ih}}^{(g)}}, \quad \tilde{f} := \frac{\mathcal{L}^{(g)} - \mathcal{L}_{\text{ih}}^{(g)}}{\mathcal{L}_{\text{ih}}^{(g)}}$$

Their Pearson correlation with the raw values is only 0.48 and 0.55 — **normalization remains open**.



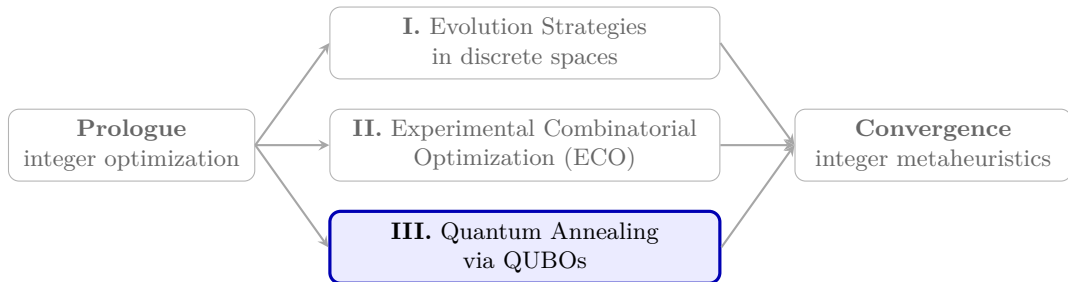
**2. Coverage:** 64 of the  $10 \cdot 9 \cdot 3 = 270$  feasible categorical combinations were visited (red stars)

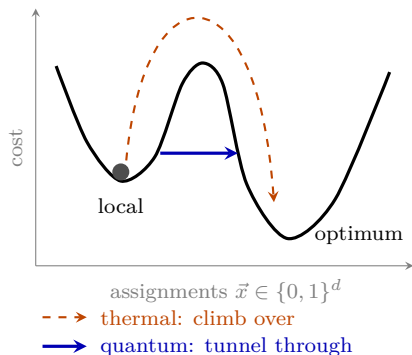
**Next:** Pareto protocols • response surfaces • transfer between crops  
• one-shot optimization • image-based objectives

## part II: take-home

- experiments are legitimate — and demanding — objective functions
- small-population, self-adaptive ES fit tiny budgets and batch-parallel rounds
- cucumbers: **77 experiments** per system, in a space of  $\sim 10^6$  protocols, sufficed to improve on the best human practice

## roadmap



quantum annealing for  $\mathcal{NP}$ -hard problems

**The target.** QUBO,  $\min_{\vec{x} \in \{0,1\}^d} \vec{x}^\top \mathbf{Q} \vec{x}$ :  $\mathcal{NP}$ -hard, and every problem in  $\mathcal{NP}$  reduces to it. Physically, its optimum is the *ground state* of an **Ising** spin system.

**The idea.** Start in the ground state of an **easy** Hamiltonian and slowly morph it into the problem Hamiltonian  $H_P$ , whose energy is the cost.

**The promise.** Evolve slowly enough and the system *stays* in the ground state, ending in the **optimum** (adiabatic theorem); it tunnels through barriers that trap local search.

The promise is not  $\mathcal{P} = \mathcal{NP}$ : “slowly enough” is set by the minimum **spectral gap**, which may close exponentially. It is a new heuristic — a physical sampler of low-cost solutions.

# the promise, the potential — and the fine print

## Evidence on hardware (4,400+ qubits, on the cloud)

- **2016:**  $\sim 10^8 \times$  faster than simulated annealing, on instances crafted with tall and thin barriers
- **2023:** coherent annealing of 5,000-qubit spin glasses; the energy drops faster with anneal time than under simulated (quantum) annealing
- **2025:** a scaling advantage in *approximate* optimization over the best classical heuristic, with 1,300 error-suppressed logical qubits

## The fine print

- the instances are shaped to the hardware graph
- classical baselines keep improving
- encoding and embedding costs sit outside the theory
- no proven speedup for general integer programs

Promising where the problem already looks like the hardware; open where it must be compiled — and that is where integer problems live.

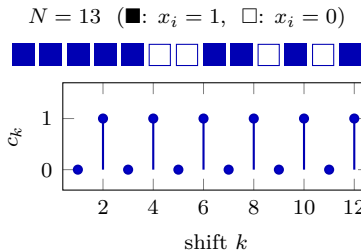
## use case: LABS as an integer program

Low autocorrelation binary sequences — a **quartic**:

$$\min_{\vec{s} \in \{-1, +1\}^N} E(\vec{s}) = \sum_{k=1}^{N-1} \left( \sum_{i=1}^{N-k} s_i s_{i+k} \right)^2$$

With  $c_k$  for the inner sums and  $s_i = 2x_i - 1$ :

$$\begin{aligned} \min \quad & \sum_{k=1}^{N-1} c_k^2 \\ \text{s.t.} \quad & c_k = \sum_{i=1}^{N-k} (2x_i - 1)(2x_{i+k} - 1), \quad k = 1, \dots, N-1 \\ & x_i \in \{0, 1\}, \quad c_k \in \{-(N-k), \dots, N-k\} \end{aligned}$$



Casting the autocorrelations as constraints unveils the **integers**:  $N$  binaries and  $N - 1$  bounded integers. The objective is convex; all the hardness sits in the quadratic equalities.

## LABS: tiny, hard, and a test for quantum optimization

| solver                    | kind       | time to solution           | evidence                           |
|---------------------------|------------|----------------------------|------------------------------------|
| MIP solver, on this model | exact      |                            | optimum proven up to $N = 41$ only |
| tailored branch and bound | exact      | $\sim 1.73^N$              | optima proven for $N \leq 66$      |
| memetic tabu search       | heuristic  | $\sim 1.34^N$              | degrades beyond $N \approx 200$    |
| QAOA (+ min. finding)     | gate model | $\sim 1.46^N$ ( $1.21^N$ ) | noiseless simulation, $N \leq 40$  |
| quantum-enhanced tabu     | hybrid     | $\sim 1.24^N$              | simulation, $N = 27\text{--}37$    |

- **one instance per**  $N$ , no instance selection — a benchmark in IOHprofiler (PBO, F18) and QOBLIB
- quantum figures are fits to classical simulations, at sizes exact search has closed; for **annealers**, the quartic energy turns  $N$  spins into  $\sim N^2/2$  QUBO variables

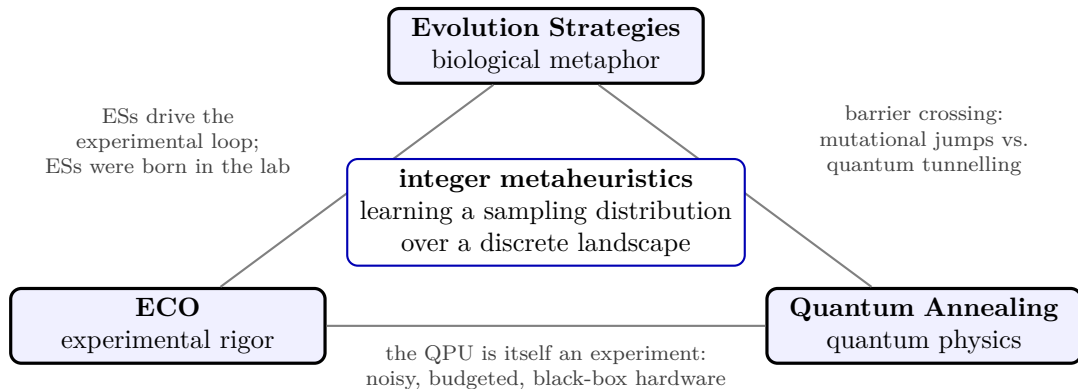
A meaningful test case: tiny, hard, and not hardware-shaped. *Under study in our group.*

Koch et al., 2025 • Packebusch & Mertens, 2016 • Shaydulin et al., 2024 • Cadavid et al., 2025

## part III: take-home

- QUBO/Ising is a common language between integer optimization and quantum hardware
- **relevance:** an annealer is one more heuristic sampler, with a new move set
- **potential:** scaling advantages on hardware-shaped instances; for integer problems, the bottleneck is the compilation — encoding and embedding — not the anneal
- **minor embedding is not so minor:** mapping the problem graph onto the hardware graph is a hard combinatorial problem of its own — it decides feasibility and multiplies the qubits (Lavi, Moraglio and Shir, 2026)
- **LABS:** a tiny, hard, quartic test case for exactly that gap

## three paths, one landscape



# conclusions

- ① Integer optimization is alive: exact theory advances, yet stays exponential — heuristics are the workhorses.
- ② **ES**: integer-native search distributions with principled (natural-gradient) updates — **INES** derives its step-size adaptation from first principles and competes with the rounding-based state of the art.
- ③ **ECO**: when the objective is an experiment, the algorithm must be budget-, noise- and batch-aware — **77 experiments** per system improved on the best human practice, and prolonged cucumber storage up to 63 days.
- ④ **QA**: QUBO maps integer problems onto physics — a new heuristic sampler, not  $\mathcal{P} = \mathcal{NP}$ ; the frontier is the compilation (encoding, minor embedding), with LABS as our test case.
- ⑤ The future of integer metaheuristics lies where the three meet.

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thank you

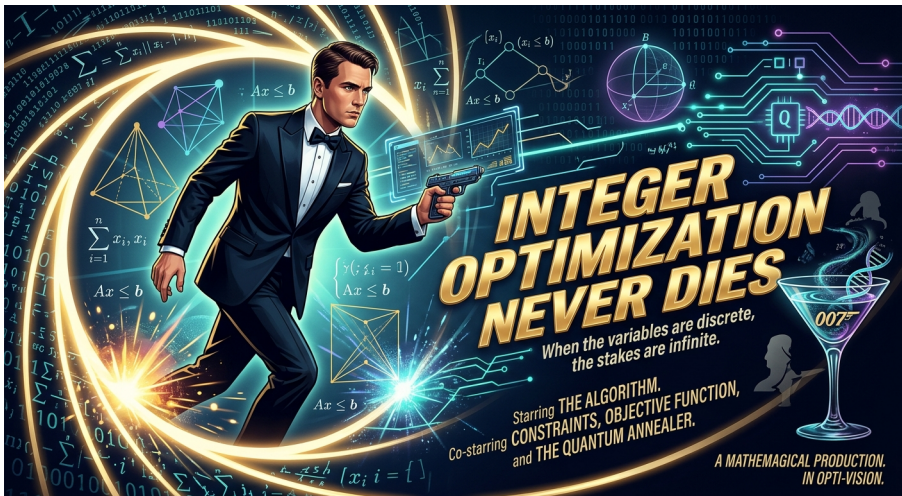


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